

MUHAMMET EMİN AYHAN

AI & Computer Vision Engineer · UAV Navigation & Perception

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SUMMARY

Computer-vision and sensor-fusion engineer focused on autonomous aerial systems and GPS-denied navigation. Hands-on experience with visual-inertial pipelines — pairing camera-based visual odometry with IMU/GNSS Kalman/EKF fusion to estimate UAV position and velocity where GPS is degraded or jammed. Comfortable taking perception from prototype to onboard deployment with C++, ROS2 and NVIDIA Jetson, on a strong Python ML/DL foundation. Eager to build navigation and autonomy for next-generation unmanned platforms.

CORE SKILLS

Languages: Python, C++, Java, Dart, Swift, C#

Navigation & Perception: Visual Odometry, SLAM, Sensor Fusion, Kalman / Extended Kalman Filter, State Estimation, Vehicle & Object Detection, Object Tracking, Image Processing

Machine Learning: Deep Learning (CNNs), Model Training & Evaluation, Data Analysis; OpenCV, NumPy, scikit-learn, PyTorch / TensorFlow

Robotics & Embedded: ROS / ROS2, NVIDIA Jetson, STM32, Real-Time Systems, Linux

Engineering & Tools: Git / GitHub, Docker, REST APIs; Spring Boot & PostgreSQL (secondary)

EXPERIENCE

AI / ML Engineer Intern

Jan 2026 – Present

AAD Bilişim ve Danışmanlık — İstanbul, Türkiye

- Developed state-estimation algorithms on IMU and sensor data using Kalman / Extended Kalman filtering for accurate position and velocity estimates under noisy, GPS-denied conditions; validated in simulation.
- Built data-processing and ML prototyping pipelines in Python for data-driven decision-support systems.
- Supported system integration of AI components into application prototypes.

Backend Developer Intern (Java Spring Boot)

Summer 2026

Enoca Bilişim — Konya, Türkiye

- Developed RESTful APIs on a layered Controller–Service–Repository architecture with Spring Boot; modeled data in PostgreSQL using JPA/Hibernate.
- Implemented role-based access control and secure authentication with Keycloak.

KEY PROJECTS

Visual-Inertial Navigation for GPS-Denied UAV Positioning

Sensor Fusion · State Estimation

- Built a Kalman/EKF sensor-fusion module fusing IMU and GNSS to estimate UAV position and velocity, engineered to stay robust when GPS is degraded or jammed.
- Ran Python simulations on noisy sensor data, improving estimation accuracy and stability over raw measurements.
- Designed to integrate with visual odometry toward a full visual-inertial navigation (VIO) solution.

Visual Odometry from UAV Imagery

TEKNOFEST Aviation AI · 2026

- Estimating UAV motion directly from onboard drone-camera frames — feature detection and matching, pose estimation — for GPS-independent positioning.
- Developing and evaluating the pipeline in Python and OpenCV on aerial footage.

Real-Time Vehicle & Object Detection

TEKNOFEST 5G & Smart Transportation · 2026

- Built deep-learning models for real-time vehicle detection and in-cabin motion / object detection from video streams.
- Focused on detection accuracy and inference suitable for real-time, edge deployment.

ML-Based CFRP Machining Optimization

TUSAŞ LIFT-UP Program

- Developed a machine-learning approach to optimize cutting parameters for machining carbon-fiber (CFRP) aerospace components.
- Performed data analysis, modeling and performance evaluation in Python, building a data-driven decision-support structure to improve quality and reduce cost.

EDUCATION

B.Sc. in Computer Engineering — Selçuk University, Konya

2022 – 2026 · GPA 3.56 / 4.00

TRAINING & SPECIALIZATIONS

Artificial Intelligence Specialization Program — National Technology Academy

Dec 2025 – Present

Autonomous Driving Technologies Specialization — National Technology Academy

Sep 2024 – May 2025

HONORS & COMPETITIONS

TEKNOFEST Air Defense Systems

Finalist · 2025

INGHub Customer Churn Prediction Datathon

Finalist · 2025

BTK Academy Hackathon

Finalist · 2026

TEKNOFEST Flying Car Simulation

6th place · 2024

CERTIFICATIONS & LANGUAGES

Deep Learning for Image Processing

BTK Academy

Deep Learning for Natural Language Processing

BTK Academy

Languages: Turkish (Native) · English (B2)